

Our derivation of the continuous-time filter starts with the previously developed discrete-time filter from Chapter 5, and then takes the limit as the time step decreases to zero. Section 8.1 shows the relationship between continuous-time white noise and discrete-time white noise, which is the foundation for the derivation of the continuous-time Kalman filter. Section 8.2 derives the Kalman filter for the case of continuous-time system dynamics and continuous-time measurements. Section 8.3 shows some creative methods to solve the continuous-time Riccati equation, which is a key component of the continuous-time Kalman filter. Section 8.4 discusses the continuous-time Kalman filter for the cases of correlated process and measurement noise, and for colored measurement noise. Section 8.5 discusses the steady-state continuous-time Kalman filter, its relationship to the Wiener filter of Section 3.4, and its relationship to linear quadratic optimal control.

8.1 DISCRETE-TIME AND CONTINUOUS-TIME WHITE NOISE

In this section, we will show the relationship between discrete-time white noise and continuous-time white noise. We need to understand this relationship because in the next section we will derive the continuous-time Kalman filter as the limiting case of the discrete-time Kalman filter as the sample time decreases to zero. First we will discuss the relationship between discrete-time and continuous-time process noise, and then we will discuss the relationship between discrete-time and continuous-time measurement noise.

8.1.1 Process noise

Consider the following discrete-time system with an identity state transition matrix and a sample time of T :

$$\begin{aligned}x_k &= x_{k-1} + w_{k-1} \\w_k &\sim (0, Q) \\x_0 &= 0\end{aligned}\tag{8.1}$$

where $\{w_k\}$ is a discrete-time white noise process. Let us see what effect the white noise has on the covariance of the state. We can solve this discrete-time system for the state as follows:

$$x_k = w_0 + w_1 + \cdots + w_{k-1}\tag{8.2}$$

The covariance of the state is therefore given as

$$\begin{aligned}E[x_k x_k^T] &= E[(w_0 + w_1 + \cdots + w_{k-1})(w_0 + w_1 + \cdots + w_{k-1})^T] \\&= E[w_0 w_0^T] + E[w_1 w_1^T] + \cdots + E[w_{k-1} w_{k-1}^T] \\&= kQ\end{aligned}\tag{8.3}$$

The value of the continuous-time parameter t is equal to the number of discrete-time steps k times the sample time T . That is, $t = kT$. We therefore see that

$$\begin{aligned}E[x(t)x^T(t)] &= E[x_k x_k^T] \\&= kQ\end{aligned}\tag{8.4}$$

The covariance of the state increases linearly with time for a given sample time T . Now consider the continuous-time system with an identity state transition matrix:

$$\dot{x}(t) = w(t) \quad (8.5)$$

where $w(t)$ is continuous-time white noise. We propose (in hindsight) the following definition for continuous-time white noise:

$$E[w(t)w^T(\tau)] = \frac{Q}{T}\delta(t - \tau) \quad (8.6)$$

where Q and T are the same as they are in the discrete-time system of Equation (8.1). $\delta(t - \tau)$ is the continuous-time impulse response; it is a function with a value of ∞ at $t = \tau$, a value of 0 everywhere else, and an area of 1. Let us compute the covariance of $x(t)$ in Equation (8.5):

$$\begin{aligned} E[x(t)x^T(t)] &= E\left[\int_0^t w(\alpha) d\alpha \int_0^t w^T(\beta) d\beta\right] \\ &= \int_0^t \int_0^t E[w(\alpha)w^T(\beta)] d\alpha d\beta \end{aligned} \quad (8.7)$$

Substituting Equation (8.6) into the above equation gives

$$\begin{aligned} E[x(t)x^T(t)] &= \int_0^t \int_0^t \frac{Q}{T}\delta(\alpha - \beta) d\alpha d\beta \\ &= \int_0^t \frac{Q}{T} d\beta \\ &= \frac{Qt}{T} \end{aligned} \quad (8.8)$$

where we have used the sifting property of the continuous-time impulse function (see Problem 4.10). Recalling that $t = kT$, we can write the above equation as

$$E[x(t)x^T(t)] = kQ \quad (8.9)$$

Comparing this with Equation (8.4), we see that the covariance of the state of the continuous-time system increases with time in exactly the same way as the covariance of the state of the discrete-time system. In other words, discrete-time white noise with covariance Q in a system with a sample period of T , is equivalent to continuous-time white noise with covariance $Q_c\delta(t)$, where $Q_c = Q/T$. Zero-mean continuous-time white noise is denoted as

$$w(t) \sim (0, Q_c) \quad (8.10)$$

which is equivalent to saying that

$$E[w(t)w^T(\tau)] = Q_c\delta(t - \tau) \quad (8.11)$$

Continuous-time white noise is counterintuitive because $w(t)$ is infinitely correlated with $w(\tau)$ at $t = \tau$, but it has zero correlation with itself when $t \neq \tau$. Nevertheless, it can be approximately descriptive of real processes. Also, continuous-time white noise is mathematically well defined and is a useful device that we will use in this chapter. Additional discussion about the relationship between discrete-time and continuous-time white noise can be found in [Kai81, Smi78].

8.1.2 Measurement noise

Now let us think about measurement noise. Suppose we have a discrete-time measurement of a constant x every T seconds. The measurement times are $t_k = kT$ ($k = 1, 2, \dots$):

$$\begin{aligned} x_k &= x_{k-1} \\ y_k &= x_k + v_k \\ v_k &\sim (0, R) \end{aligned} \quad (8.12)$$

From the Kalman filter equations in Section 5.1 we find that the *a posteriori* estimation-error covariance is given by

$$P_{k+1}^+ = \frac{P_k^+ R}{P_k^+ + R} \quad (8.13)$$

From this it can be shown that

$$\begin{aligned} P_k^+ &= \frac{P_0 R}{k P_0 + R} \\ \lim_{P_0 \rightarrow \infty} P_k^+ &= \frac{R}{k} \\ &= \frac{RT}{t_k} \end{aligned} \quad (8.14)$$

The error covariance at time t_k is independent of the sample time T if

$$R = \frac{R_c}{T} \quad (8.15)$$

where R_c is some constant. This implies that

$$\lim_{T \rightarrow 0} R = R_c \delta(t) \quad (8.16)$$

where $\delta(t)$ is the continuous-time impulse function. This establishes the equivalence between white measurement noise in discrete time and continuous time. The effects of white measurement noise in discrete time and continuous time are the same if

$$\begin{aligned} v_k &\sim (0, R) \\ v(t) &\sim (0, R_c) \end{aligned} \quad (8.17)$$

Equation (8.15) specifies the relationship between R and R_c , and the second equation above is a shorthand way of saying

$$E[v(t)v(\tau)] = R_c \delta(t - \tau) \quad (8.18)$$

8.1.3 Discretized simulation of noisy continuous-time systems

The results of the above sections can be combined with the results of Section 1.4 to obtain a discretized simulation of a noisy continuous-time system for the purpose